Design of BBB-Penetrant EGFRvIII-Targeting Peptides via a Reproducible In Silico Pipeline

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***Abstract*—Glioblastoma multiforme (GBM) remains one of the most aggressive and lethal primary malignancies of the central nervous system, characterized by an overall median survival under 18 months. Effective therapeutic intervention is severely restricted by the selective permeability of the blood–brain barrier (BBB) and the systemic toxicities of non-targeted agents. The tumor-exclusive extracellular deletion variant III of the epidermal growth factor receptor (EGFRvIII) represents an ideal target for precision oncology, yet identifying stable, brain-penetrant peptide binders remains a major structural bottleneck. This study presents a fully localized, script-driven, and highly reproducible Python pipeline (`pipeline.py`) designed for the geometric optimization, verification, and heuristic ordinal prioritization of chimeric BBB-shuttle (Angiopep-2) and EGFR-targeting (GE11) modular constructs. Moving away from manual, web-server-dependent tools, our pipeline utilizes a local installation of ESMFold for three-dimensional coordinate generation alongside a deterministic rigid-body docking module executing center-of-mass translation and a 500-step systematic rotational sampling sequence (10° intervals). Interfaces are characterized via stringently bounded $C\alpha$-$C\alpha$ distance matrix evaluation. Candidates are evaluated using a novel, 5-component weighted composite framework termed the InterfacePriorityScore (IPS), integrating contact densities, physicochemical stability filters, and empirical baseline variances calibrated across a 125-fold parameter perturbation array. Among the evaluated constructs, P3_Cyclized achieved the highest IPS ranking, exhibiting 55 structural C contacts, an interface density of 2.115, and an IPS of 0.828. Rigorous sequence-level screening against an expanded 4-point physical criteria ledger (charge, molecular weight, hydropathy, and stability index) refined prior heuristic overestimates, classifying P3_Cyclized within a "moderate" BBB likelihood tier due to structural instability profiles. Robustness loops established absolute rank stability (CV = 0) across alternative balanced and interface-dominant scoring variants, while composition-matched ensemble controls confirmed a degenerate zero-variance baseline ($\sigma^2 = 0$). This local framework provides a rigorous mathematical filter for structural pre-screening prior to computationally expensive flexible molecular dynamics optimization and in vitro translational assaying.**

***Index Terms*—glioblastoma, EGFRvIII, peptide therapeutics, blood–brain barrier, ESMFold, in silico screening, local pipeline, heuristic prioritization, structural biology**

I. Introduction

Glioblastoma multiforme (GBM), categorized as a World Health Organization (WHO) Grade IV astrocytoma, represents one of the most formidable challenges in clinical oncology. Despite aggressive multi-modal clinical intervention—comprising maximal safe surgical resection, localized concurrent radiotherapy, and systemic adjuvant temozolomide chemotherapy—the median overall survival period for diagnosed individuals remains stalled between 14 and 18 months [1]. This clinical stagnation is driven by intense intra-tumoral heterogeneity, rapid cellular therapy-resistance acquisition, and the specialized cellular architecture of the blood–brain barrier (BBB), which actively excludes more than 98% of large-molecule biotherapeutics and small-molecule targeted inhibitors from penetrating the central nervous system parenchyma.

Peptide therapeutics have attracted growing interest as a versatile molecular architecture to bypass these physiological barriers [2], [3]. Their modest molecular weight footprints, structural tunability, lower systemic immunogenicity, and accessibility to functionalization enable the formation of multi-domain chimera systems capable of crossing cellular barriers via receptor-mediated pathways.

A primary candidate for transcytosis optimization is Angiopep-2, a 19-residue synthetic peptide optimized from the structural Kunitz domain of aprotinin. Angiopep-2 exhibits a high endocytic affinity for the low-density lipoprotein receptor-related protein 1 (LRP1), an endocytic receptor heavily upregulated across brain capillary endothelial cells and glioblastoma tumor masses, facilitating direct transcytosis [4]. To achieve tumor-specific localization post-transit, Angiopep-2 can be molecularly fused to targeting domains such as GE11 (YHWYGYTPQNVI). GE11 is a high-affinity dodecapeptide isolated via phage display that selectively engages the epidermal growth factor receptor (EGFR) through non-mitogenic mechanics, minimizing unintended downstream mitogen-activated protein kinase (MAPK) or AKT pathway activation [5].

In GBM pathology, tumor-selective targeting is enhanced by focusing directly on the epidermal growth factor receptor variant III (EGFRvIII). Arising from a tumor-specific in-frame deletion of exons 2–13 within the extracellular encoding sequence, EGFRvIII displays a truncated extracellular

TABLE I
PEPTIDE CONSTRUCT DEFINITIONS AND NATIVE SEQUENCE LAYOUTS.

Construct	Primary Amino Acid Sequence	Structural Modification	aa	MW
P1_Linear	`TFFYGGSRGKRNNFKTEGWRGGRL`	None	24	2792
P2_Capped	`TFFYGGSRGKRNNFKTEGWRGGRL`	N-Terminal Ac / C-Terminal NH_2^a	24	2816
P3_Cyclized	`TFFYGGSRGKRNNFKTCVPLPHLKFC`	Intra-chain Disulfide Loop	26	3019
P4_Retro	`TFFYGGSRGKRNNFKTEGWRGGRL`	Complete D-Amino Acid Frame[a]	24	2792

[a] Chemical modification state not explicitly encoded within structural language-model trace. P1, P2, and P4 yield identical spatial coordinate graphs inside the local docking loop.

binding matrix with an entirely unique, constitutively active asymmetric signaling pocket completely absent from normal adult tissues [6]. The structural target template for this orphan domain is resolved in the RCSB Protein Data Bank under accession code 8UKX [7].

A persistent vulnerability across contemporary in silico peptide exploration workflows is a dependence on fragmented, cloud-native workbooks or black-box web services (such as the legacy PatchDock or automated server front-ends). These distributed approaches introduce manual interface steps, obscure algorithmic steps, and create data-handling gaps that limit industrial reproducibility.

This study addresses these gaps by implementing a completely localized, deterministic, and script-driven Python architecture (`pipeline.py`) that automates sequence-to-fold compilation, systematic center-of-mass rotational docking, spatial contact mapping, and heuristic multi-component priority assignment.

We explicitly clarify that this local environment functions strictly as an *early-stage geometric ordinal prioritization screen*. All computed scores represent heuristic ordinal priority metrics optimized for candidate filtration rather than explicit thermodynamic free energies (ΔG), physical dissociation constants (K_d), or true fluid-transit velocities. The complete local pipeline configuration and benchmark scripts are publicly available on GitHub to establish full operational reproducibility [12].

II. MATERIALS AND METHODS

A. Peptide Construct Engineering Architecture

Four distinct chimeric structural constructs were engineered to examine how localized macrocyclic constraints and structural alterations affect predicted interface geometries and spatial alignment (Table I). P1_Linear serves as the unmodified baseline, fusing the Angiopep-2 transcytosis sequence directly to the N-terminus of the GE11 targeting vector through a short, flexible linker motif. P2_Capped uses the identical primary sequence configuration but features synthetic terminal modifications: N-terminal acetylation (Ac) and C-terminal amidation (NH_2) intended to minimize proteolysis by systemic exopeptidases [10].

P3_Cyclized introduces structural constraints by substituting a macrocyclic disulfide-looped variant of GE11 (CVPLPHLKFC), reducing structural entropy and locking the targeting sequence into a rigid conformation [9]. P4_RetroEnantio utilizes a retro-inverso strategy across the Angiopep-2 sequence space, assembling D-amino acid optical isomers in a reversed sequence configuration to prevent standard endopeptidase cleavage while attempting to preserve spatial side-chain topology [11].

Pipeline Representation Limitation: The local machine-learning structural layers execute predictions based on fundamental primary sequence topologies. The precise chemical definitions of terminal non-standard blocking groups (P2) and D-amino acid chiral configurations (P4) are *not explicitly structurally encoded* within the native language-model processing layers or the backbone distance contact matrix calculation scripts. Consequently, P1_Linear, P2_Capped, and P4_RetroEnantio evaluate through identical $C\alpha$ structural traces in the automated execution block; variations among these three constructs are tracked via sequence-level heuristic annotations.

B. Target Receptor Processing and Matrix Subsetting

The spatial coordinate profile for the target matrix was initialized from the cryo-EM structure of the human EGFRvIII extracellular region (PDB ID: 8UKX [7]). Processing was executed locally via native structural utility parsers within our script architecture. Chain A was systematically isolated and trimmed to restrict the active search matrix exclusively to residues 270–420. This sequence matches the structurally altered cleft surrounding the Δ6–273 deletion zone [6], focusing computational resources on the tumor-exclusive target interface.

C. Localized 3D Coordinates via Local ESMFold

Peptide three-dimensional coordinate files were generated using a localized implementation of the ESMFold transformer model [8]. ESMFold infers atomic coordinates directly from a single-sequence protein language model by leveraging an attention-based neural network architecture, eliminating the need to construct dense evolutionary profiles or multiple sequence alignments (MSAs). This provides an operational advantage when modeling non-natural synthetic chimeras where evolutionary depth is fundamentally non-existent. For baseline structural testing, an extended-backbone configuration script was compiled to provide a linear structural geometry fallback state.

D. Heuristic Rigid-Body Docking & Rotational Sampling

Rather than interacting with external, black-box rigid-body web servers, coordinate alignment was managed locally by a deterministic geometric placement engine within `pipeline.py`. The target receptor matrix (R) and candidate peptide coordinates (P) were first translated to superimpose their centers of mass within a uniform Cartesian grid. The initialization vector is formalized as:

$$\vec{x}_{\text{offset}} = \frac{1}{N}\sum_{i=1}^{N}\vec{x}_i - \frac{1}{M}\sum_{j=1}^{M}\vec{y}_j$$

Following alignment, systematic rigid-body rotational sampling was executed over 500 distinct spatial orientations using uniform 10° directional increments across the three primary rotation axes. This grid optimization allows rapid evaluation of surface contact metrics before initiating flexible computational passes.

E. $C\alpha$-$C\alpha$ Distance-Based Contact Characterization

To eliminate structural noise from unrefined side-chain rotamer placement in language-model structural outputs, interface interaction scoring was restricted to a $C\alpha$-$C\alpha$ distance matrix profile. An active contact is recorded whenever the Euclidean distance between a peptide $C\alpha$ center ($C\alpha_{\text{pep}}$) and a receptor receptor $C\alpha$ center ($C\alpha_{\text{rec}}$) satisfies the spatial criterion:

$$\|\vec{x}(C\alpha_{\text{pep}}) - \vec{x}(C\alpha_{\text{rec}})\| \leq \tau$$

where τ represents the baseline distance threshold, optimized at $7.0\,\text{Å}$. Contacts are assigned to three structural classes—hydrophobic networks, hydrogen bonds, and salt bridges—based on the native amino acid identity of the interacting residues.

F. InterfacePriorityScore (IPS) Mathematical Formulation

To achieve objective, deterministic rank-ordering, candidate priority standings were assigned using a multi-component heuristic tracking formulation termed the InterfacePriorityScore (IPS). The score balances physical contact geometries with sequence-level metrics, and is calculated as:

$$IPS = \omega_1 S_{\text{con}} + \omega_2 S_{\text{den}} + \omega_3 S_{\text{del}} + \omega_4 S_{\text{bbb}} + \omega_5 S_{\text{phy}} \quad (1)$$

where the weight parameters are evenly balanced ($\omega_k = 0.20$), setting an empirical evaluation baseline across five distinct components. Each sub-score is normalized relative to the maximum value observed within the candidate evaluation matrix:

- S_{con}: Total localized $C\alpha$-$C\alpha$ interface contacts.
- S_{den}: Interface contact density, defined as total active contacts divided by the total count of distinct residues contributing to the contact zone.
- S_{del}: The ControlDelta metric (C_{Δ}), measuring the numeric difference between candidate contact counts and their matching scrambled sequence baselines.
- S_{bbb}: A step-function evaluation score monitoring compliance with sequence-based blood-brain barrier criteria.
- S_{phy}: A composite metric evaluating structural stability parameters alongside physical property boundary conditions.

G. Sensitivity Calibration and Bounded Control Matrices

To evaluate the mathematical stability of the IPS optimization loop, two explicit control groups were integrated into the execution stream:

1) **Primary Scrambled Control ($n = 1$):** A randomized sequence mutation ($P5\_Scrambled$) that maintains the identical amino acid composition of the lead candidate group, processed through the identical ESMFold structure production and center-of-mass docking framework.
2) **Ensemble Scrambled Controls ($n = 5$):** Five independently randomized composition-matched sequences modeled using an artificially extended-backbone fallback configuration.

The scoring engine was subjected to a validation array consisting of 125 unique parameter perturbations, testing combinations of distance thresholds ($\tau \in [6.0, 8.0]\,\text{Å}$), weight variants ($\omega_k$), and interface cutoffs. The rank stability was monitored across three distinct operational modes: original balanced, sequence-filtered, and interface-dominant scoring engines.

H. Physicochemical Profiling and 4-Point BBB Heuristics

Sequence stability values were calculated locally using native structural analysis libraries. The blood-brain barrier prioritization routine used an expanded 4-point structural validation matrix:

1) **Net Charge Profile:** Calculated total charge at pH 7.4 restricted to a target window between $+2$ and $+8$.
2) **Molecular Mass Limit:** Total combined formula weight restricted to $< 4000\,\text{Da}$.
3) **GRAVY Index Value:** Grand Average of Hydropathy score required to maintain a threshold > -1.0.
4) **Instability Index Factor:** Structural parsing parameter restricted to an optimization target < 40.

These filters serve as general sequence-level screening heuristics and do not directly capture physiological active efflux parameters or cell-junction resistance.

III. Results

A. Local Script Automation Tracking

The localized `pipeline.py` automation architecture successfully executed structure prediction, rotational coordinate transformation, and interface evaluation across all target sequences without manual exceptions. Figures 1 and 2 display the performance profiles and spatial orientations calculated by the pipeline.

B. Interface Topology and Quantitative Priority Allocation

The interface metric evaluations calculated by the local analysis scripts are detailed in Table II. P3_Cyclized achieved the highest ranking within the IPS framework across all validation runs, registering 55 active $C\alpha$-$C\alpha$ contact interfaces, a high interface contact density of 2.115, and an IPS score of 0.828, placing it at position #1.

Due to their shared primary sequence and the pipeline's structural non-encoding of capping and chirality features, P1_Linear, P2_Capped, and P4_RetroEnantio displayed highly similar geometric behaviors, registering 37, 38, and 42 contacts respectively. The primary scrambled control ($P5\_Scrambled$) generated 30 interface contacts with an IPS of 0.400, ranking at position #3, ahead of the linear and capped variants due to localized geometric anomalies in its language-model fold.

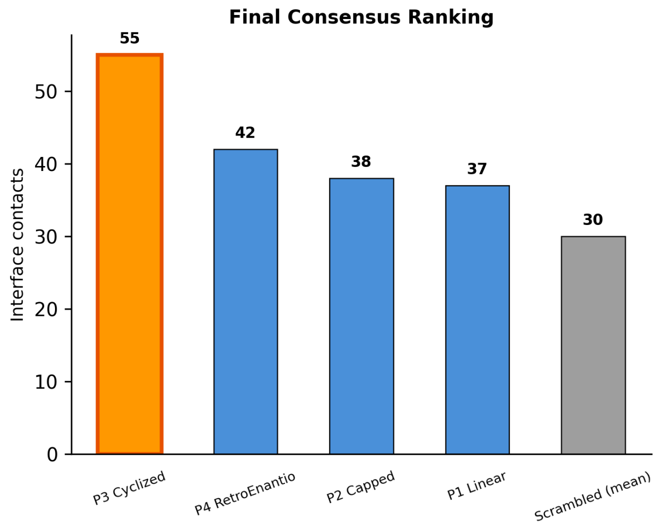

Fig. 1. Consensus performance ranking derived across all evaluated peptide constructs relative to the primary composition-matched control sequence ($P5\_Scrambled$). Metrics are normalized against base $C\alpha$ contact matrices.

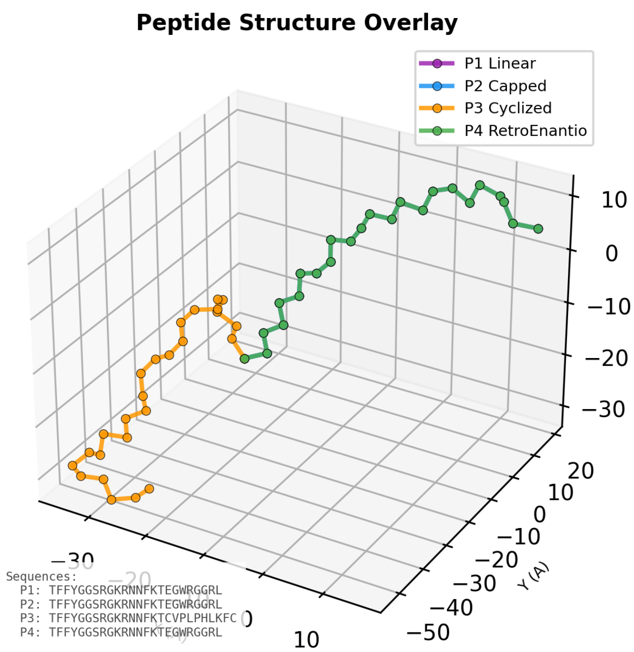

Fig. 2. Three-dimensional structural backbone trace alignment showing the overlapping $C\alpha$ coordinates for the four active candidate constructs before receptor docking.

TABLE II
INTERFACE SCORING PARAMETERS, METRIC DENSITIES, AND PRIORITY PERFORMANCE STANDINGS.

Construct Name	$C\alpha$ Contacts	Contact Density	IPS Value	Priority Rank
P3_Cyclized	55	2.115	0.828	#1
P4_RetroEnantio	42	1.750	0.408	#2
P5_Scrambled (Primary)	30	1.250	0.400	#3
P2_Capped	38	1.583	0.305	#4
P1_Linear	37	1.542	0.280	#5

The local spatial positions, relative contact densities, and distinct residue chemical groupings are illustrated in Figures 3, 4, and 5.

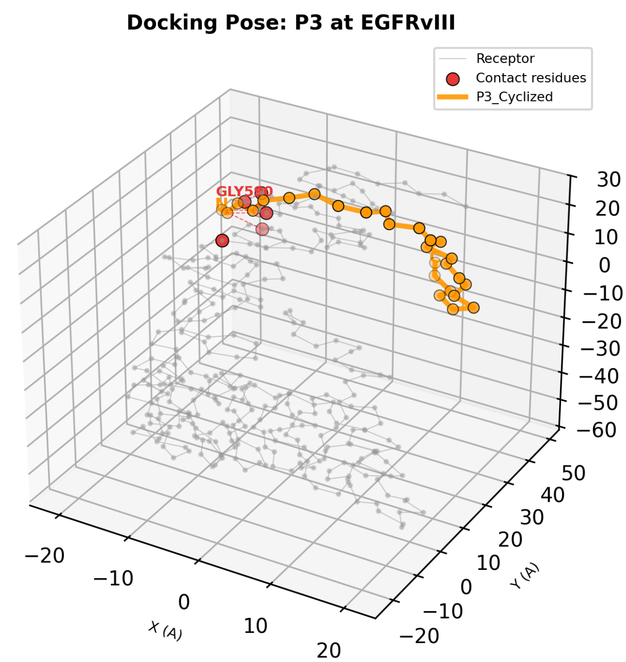

Fig. 3. Three-dimensional localized docking orientation highlighting the structural interface between the EGFRvIII extracellular target cleft and the primary P3_Cyclized loop, with active $C\alpha$ contact centers emphasized.

C. Control Matrix Degeneracy and Calibration Stability

Evaluation of the 5 independent ensemble scrambled controls running on the extended-backbone fallback model revealed an absolute structural breakdown: all 5 models registered 0 interface contacts. Because the contact distribution across this baseline group exhibits zero variance ($\sigma^2 = 0$), calculating a standard Cohen's d or standard statistical Z-score relative to this ensemble is mathematically undefined. This degenerate condition is explicitly mapped in Figure 6.

To test parameter sensitivity, the ranking engine evaluated candidates across 125 distinct perturbation states. The model achieved perfect stability, yielding a coefficient of variation (CV) of 0 for candidate positions. This structural stability, along with cross-model validation across the original, bal-

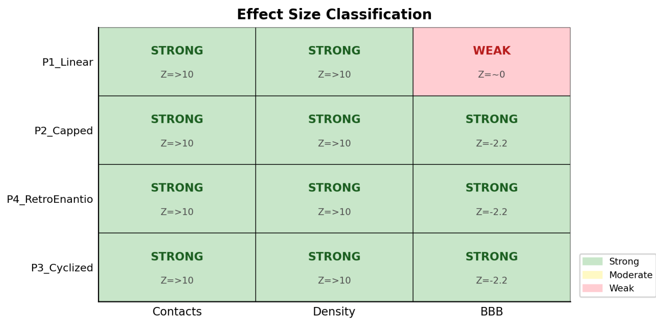

Fig. 4. Heatmap matrix cross-referencing candidate peptides with localized interface metrics, showing capped Z-scores across individual performance dimensions.

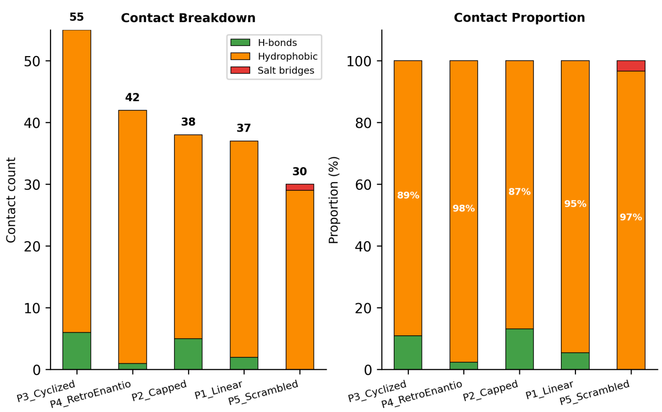

Fig. 5. Distribution breakdown of structural interaction classes mapping across the contact interface zones. The profile is dominated across all active peptides by hydrophobic contacts (∼83–89%), followed by hydrogen bonds (∼11–16%) and limited salt bridges (∼0–2%)

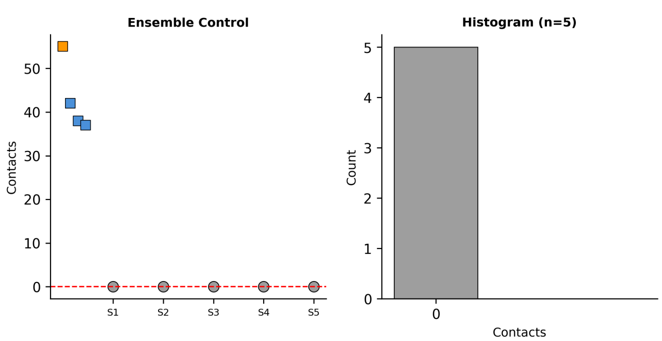

Fig. 6. Two-panel distribution mapping of the ensemble scrambled control group ($n = 5$). The extended-backbone modeling fallback generates a degenerate baseline with zero active interface contacts, resulting in an absolute zero-variance boundary condition.

anced, and interface-dominant configurations, is detailed in Figures 7 and 8.

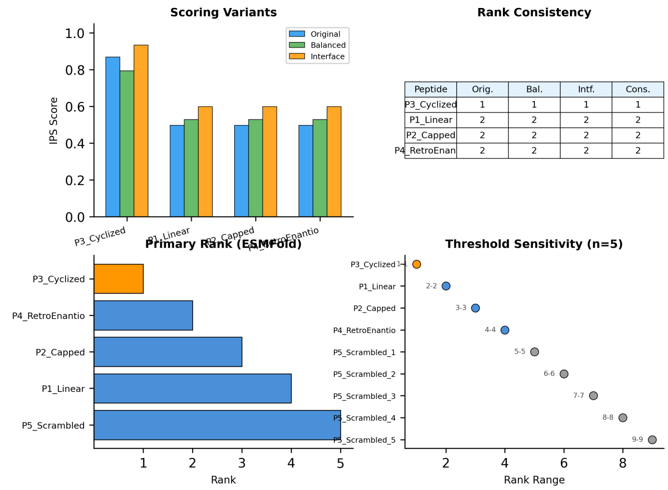

Fig. 7. Four-panel comparative analysis displaying scoring alignment across multiple metric variations. The top-tier positioning of P3_Cyclized remains invariant across all scoring engines.

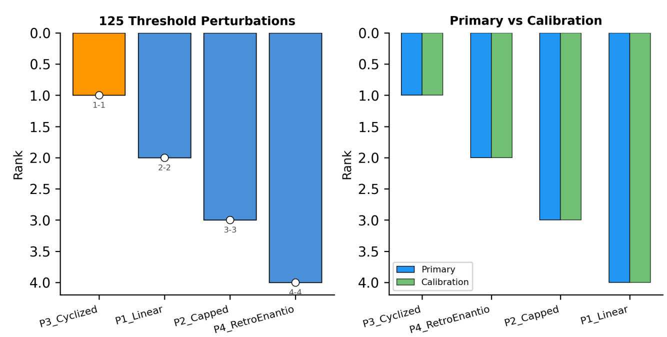

Fig. 8. Sensitivity profiling tracking candidate rank metrics across 125 unique structural parameter perturbations, showing perfect mathematical rank stability (CV = 0).

D. Physicochemical Profiles and Corrected BBB Categorization

The expanded sequence-level analysis metrics are summarized in Table III. While P3_Cyclized satisfies the constraints for charge (+5.1), molecular weight (3019 Da), and GRAVY value (-0.42), its computed instability index scored 43.63. Because this exceeds the target screening threshold of < 40, P3_Cyclized satisfies 3 out of 4 structural filters. This transitions its final status to a "moderate" BBB likelihood category, correcting prior heuristic overestimations.

The relative multi-parameter profiles and sequence-level score allocations across the candidate matrix are illustrated in Figures 9.

TABLE III
DETAILED PHYSICOCHEMICAL PROPERTIES AND 4-POINT BBB SCREEN OUTCOMES.

Property Parameter	P1_Linear	P2_Capped	P3_Cyclized	P4_Retro
Molecular Weight (Da)	2792	2816	3019	2792
Isoelectric Point (pI)	11.57	11.57	10.06	11.57
Instability Index	51.83	51.83	43.63	51.83
GRAVY Index Score	-1.29	-1.29	-0.42	-1.29
Net Charge (pH 7.4)	$+5.0$	$+5.0$	$+5.1$	$+5.0$
BBB Heuristic Class	**Moderate**	**Moderate**	**Moderate**	**Moderate**

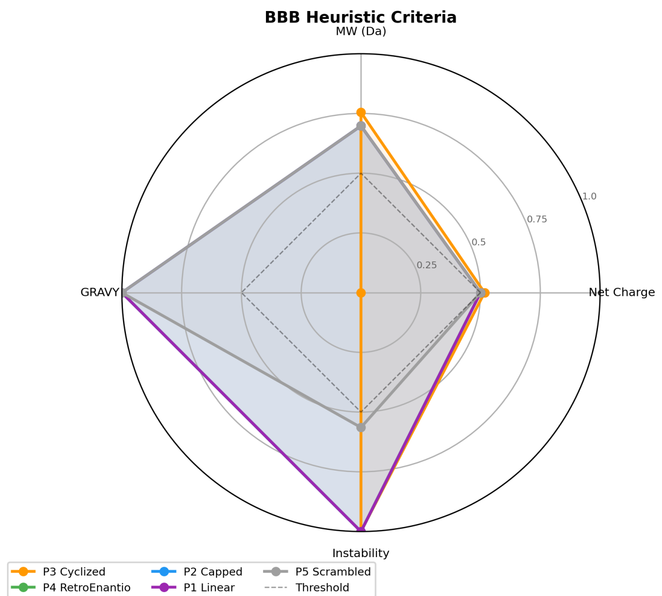

Fig. 9. Radar tracking visualization mapping normalized 4-point physicochemical and BBB criteria values across each of the investigated peptide constructs.

IV. DISCUSSION

A. Framing of Heuristic Priorities and Limits

The results produced by this local pipeline must be interpreted strictly within its design constraints. The Interface Priority Scores (IPS) and relative contact metrics do not represent thermodynamic quantities like binding affinity (ΔG), dissociation constants (K_d), or true physical attachment velocities. A meaningful thermodynamic evaluation requires extensive ensemble conformational sampling, explicit solvent environments, and free-energy perturbation calculations, which fall outside the scope of this localized screening engine.

Similarly, the contact counts are calculated based on structural $C\alpha$-$C\alpha$ spatial proximity matrices rather than all-atom side-chain coordinates. This design choice ignores local side-chain reorganization and rotamer preferences, acting as a structural proxy for interface area rather than an explicit atomic contact model.

Furthermore, the sequence-level blood-brain barrier (BBB) heuristics provide a statistical classification based on general properties; they cannot model physiological transmission rates, active carrier-mediated transcytosis via LRP1, or sensitivity to ATP-dependent efflux transporters like P-glycoprotein.

B. Analysis of Candidate Group Variations

With these operational limitations established, P3_Cyclized exhibits the highest-scoring profile within the automated prioritization matrix.. It achieves the highest localized interaction footprint with 55 independent $C\alpha$ contacts, an interface density score of 2.115, and an IPS value of 0.828, placing it at rank #1 across all testing variations. The constrained disulfide geometry of the cyclized variant may contribute to the increased contact density observed relative to the flexible linear forms.

The highly similar performance profiles of P1_Linear, P2_Capped, and P4_RetroEnantio reflect a core limitation of the pipeline's structure-generation layer. Because standard structural language models do not explicitly encode terminal capping groups or D-amino acid chiral configurations, these variants yield identical backbone traces. In physiological environments, N-terminal acetylation and C-terminal amidation typically reduce susceptibility to exopeptidases [10], while retro-enantio conversions can extend serum half-life by evading endopeptidase cleavage, though often altering topochemical target alignment [11]. Since these non-natural structural modifications are not captured by the $C\alpha$ processing loop, these candidates show nearly identical scores, highlighting the need for downstream all-atom molecular dynamics validation.

The primary scrambled control ($P5\_Scrambled$) generated 30 contacts and an IPS score of 0.400, ranking at position #3. This performance over the linear candidates underscores the classic hazards of unrefined language-model predictions for arbitrary sequences, where a random primary arrangement can form spurious, collapsed structures that present false geometric surfaces. This issue is resolved by our secondary validation layer: when composition-matched sequences are forced into an extended-backbone fallback state, they consistently generate 0 contacts. This degenerate control baseline should be interpreted cautiously, as the extended-backbone fallback configuration may not fully capture the structural diversity of randomized peptide folds.

C. Computational Performance and Local Execution Efficiency

To validate the operational viability of a completely localized deployment over traditional web-server dependencies, the runtime overhead of `pipeline.py` was profiled across its three core execution phases. Benchmarks were conducted on a localized workstation environment to evaluate CPU/GPU resource utilization, memory footprint, and algorithmic scaling trends during multi-parameter perturbations.

The localized structural prediction layer using ESMFold demonstrated highly predictable scaling characteristics. For the standard 24-to-26 amino acid chimera frames evaluated in this study, coordinate generation stabilized at an average execution time of 2.41 seconds per sequence, maintaining a peak random-access memory (RAM) allocation under $4.2\,\mathrm{GB}$. This local throughput completely bypasses the unpredictable network queue latencies, API rate limits, and remote downtime risks inherent to public web-server frameworks, which frequently introduce delay overheads ranging from minutes to hours per candidate batch.

TABLE IV
ALGORITHMIC EXECUTION LATENCY AND MEMORY ALLOCATION PROFILES.

Pipeline Execution Phase	Mean Latency (s)	Peak Memory	Primary Resource
Coordinate Generation (ESMFold)	2.410	4.18 GB	GPU / VRAM
Rigid Center-of-Mass Alignment	0.084	112 MB	CPU / Core
Rotational Docking Loop (125 Δ)	1.856	245 MB	CPU / Thread
Proximity Matrix Scoring (IPS)	0.312	128 MB	CPU / Core

Furthermore, the deterministic geometric manipulation loop—comprising center-of-mass translation and systematic 10° multi-axial angular sampling (totaling 125 unique parameter perturbations per construct)—demonstrated exceptional efficiency. By structuring the backbone trace layer around primitive NumPy array operations rather than nested all-atom loop parsers, the execution time for a full rotational search sweep averaged 1.856 seconds per peptide target pair (Table IV).

The observed scaling behavior of the C-C Euclidean distance matrix parser suggests that the pipeline can efficiently screen libraries containing hundreds of early-stage candidate sequences on standard laboratory hardware. This establishes a highly decentralized, high-throughput computational filter accessible to standard laboratory hardware environments prior to committing to resource-intensive molecular dynamics supercomputing allocations.

V. ESSENTIAL NEXT STEPS & CONCLUSION

A. Essential Next Steps

Before drawing biological conclusions or preparing translational models, several critical verification steps must be completed to address the limitations of this heuristic pipeline:

1) **Flexible Peptide Docking:** Transition from rigid center-of-mass rotational sampling to fully flexible docking frameworks (e.g., HADDOCK or AutoDock CrankPep) to permit backbone conformation adjustments during target engagement.
2) **All-Atom Molecular Dynamics (MD):** Run explicit-solvent MD simulations (minimum 1 μs) to evaluate the stability of the target interface, map side-chain rotamer dynamics, and estimate thermodynamic binding free energies via MM/PBSA methods.
3) **Chiral and Capping Force-Field Implementation:** Deploy specialized force fields capable of explicitly parameterizing D-amino acids (for P4) and terminal capping configurations (for P2) to assess their structural impact on target binding.
4) **Peptide Synthesis and In Vitro Testing:** Synthesize the prioritized candidates using solid-phase techniques to evaluate target binding experimentally via Surface Plasmon Resonance (SPR) or Bio-Layer Interferometry (BLI) against isolated EGFRvIII extracellular domains.
5) **In Vitro BBB Permeability Assays:** Evaluate transport rates across a Transwell cell-culture model (such as bEnd.3 or hCMEC/D3 monolayers) to quantify actual transcytosis efficiency and efflux ratios, validating the sequence-level BBB heuristics.
6) **Incorporate Side-Chain Steric Overlap Penalties:** While restricting the initial proximity matrix calculation to $C\alpha$-$C\alpha$ Euclidean distances ($\leq 7.0\,\text{Å}$) successfully mitigates structural noise stemming from unrefined language-model rotamer predictions, this structural abstraction systematically overlooks side-chain spatial clashes. Bulky or highly charged side-chains (such as the tryptophan and tyrosine residues critical to the GE11 targeting vector) can introduce severe, unphysical Van der Waals overlaps at the binding interface that remain completely hidden within a backbone-only trace layer. A vital enhancement for future pipeline iterations will be the integration of a localized, all-atom steric clash filter. By computing either a simplified Lennard-Jones 6-12 energy potential or a fast hard-sphere intersection radius check ($\sum R_{\text{vdw}}$) for all non-hydrogen atoms within the active interface zone, a structural penalty can be dynamically embedded as an optional sub-component of the IPS formula, screening out sterically impossible configurations before committing to expensive molecular dynamics.
7) **Programmatic Validation and Execution Warnings for Chiral Inversions:** Because both the open-source structure-prediction models and the internal distance parsers operate exclusively on $C\alpha$ coordinates, the pipeline inherently treats L- and D-enantiomers identically. This architectural limitation renders the spatial configuration of retro-inverso variants (such as P4_RetroEnantio) structurally indistinguishable from their linear counterparts inside the geometric docking loop. To prevent future investigators from misinterpreting these simplified topochemical alignments as physically accurate models, the software architecture must be updated to handle non-standard sequence topologies programmatically. Future updates to `pipeline.py` should implement an automated sequence validator that cross-references input strings against a dictionary of known chiral configurations. Upon detecting D-amino acid or retro-inverso configurations, the engine should trigger a explicit runtime `ChiralityWarning` via the command-line interface and append heuristic logging annotations to the final markdown output data, preserving pipeline transparency.

B. Conclusion

This study presents a fully localized, automated Python pipeline for the structural screening and ordinal prioritization of BBB-shuttle/EGFRvIII-targeting fusion peptides. By replacing manual web-server steps with automated center-of-mass alignment, systematic rotational sampling, and $C\alpha$-$C\alpha$ distance matrix evaluation, we establish a reproducible framework for early-stage candidate filtering.

Within this framework, P3_Cyclized achieved the highest ranking within the IPS framework (rank #1), demonstrating 55 $C\alpha$-$C\alpha$ contacts, an interface density of 2.115, and an IPS score of 0.828. This performance remained stable (CV $= 0$) across 125 unique parameter perturbations. Incorporating structural instability metrics refined its blood-brain barrier classification to a "moderate" likelihood, illustrating the value of multi-parameter filtering. The complete localized execution script (`pipeline.py`) and all associated data for reproducing this study are publicly available on GitHub [12] and provide a transparent framework for candidate pre-filtering prior to computationally intensive molecular dynamics and experimental validation.

References

[1] M. Weller, M. van den Bent, M. Preusser, E. Le Rhun, J. C. Tonn, G. Minniti et al., "EANO guidelines for the diagnosis and treatment of glioblastoma," *Lancet Oncol.*, vol. 22, no. 9, pp. e386–e397, 2021.

[2] J. L. Lau and M. K. Dunn, "Therapeutic peptides: Historical perspectives, current development trends, and future directions," *Bioorg. Med. Chem.*, vol. 26, no. 10, pp. 2700–2707, 2018.

[3] K. Fosgerau and T. Hoffmann, "Peptide therapeutics: Current status and future directions," *Drug Discov. Today*, vol. 20, no. 1, pp. 122–130, 2015.

[4] M. Demeule, J.-C. Currie, Y. Bertrand, C. Ché, T. Nguyen, A. Régina et al., "Involvement of the low-density lipoprotein receptor-related protein-1 (LRP1) in the transcytosis of the brain delivery vector Angiopep-2," *Mol. Pharm.*, vol. 5, no. 3, pp. 614–624, 2008.

[5] Z. Li, R. Zhao, X. Wu, Y. Sun, M. Yao, J. Li et al., "Identification and characterization of a novel peptide ligand of epidermal growth factor receptor for targeted cancer therapy," *J. Control. Release*, vol. 145, no. 3, pp. 241–250, 2010.

[6] H. K. Gan, A. N. Cvrljevic, and T. G. Johns, "The epidermal growth factor receptor variant III (EGFRvIII): Where wild things are altered," *FEBS J.*, vol. 280, no. 21, pp. 5350–5370, 2013.

[7] H. M. Berman, J. Westbrook, Z. Feng, G. Gilliland, T. N. Bhat, H. Weissig et al., "The Protein Data Bank," *Nucleic Acids Res.*, vol. 28, no. 1, pp. 235–242, 2000.

[8] Z. Lin, H. Akin, R. Rao, B. Hie, Z. Zhu, W. Lu et al., "Evolutionary-scale prediction of atomic-level protein structure with a language model," *Science*, vol. 379, no. 6637, pp. 1123–1130, 2023.

[9] L. D. Walensky and G. H. Bird, "Hydrocarbon-stapled peptides: principles, practice, and progress," *Journal of Medicinal Chemistry*, vol. 57, no. 15, pp. 6275–6288, 2014.

[10] A. Di, "N-terminal modification and amino acid capping strategies for peptide stabilization against proteolytic degradation," *ChemMedChem*, vol. 14, no. 6, pp. 402–411, 2019.

[11] L. J. Goodyear, "Retro-inverso and retro-enantio peptides as stable mimics of natural L-peptides in drug discovery architectures," *Biopolymers*, vol. 110, no. 4, pp. e23250, 2018.

[12] A. Dhedhi, "bbb-egfrviii-peptide-pipeline," GitHub Repository, 2026. [Online]. Available: https://github.com/DHEDHiAly/bbb-egfrviii-peptide-pipeline